

Answers to the Q. No-01

$$F = k \cdot \frac{q_1 q_2}{r^2}$$

$$\Rightarrow r^2 = \frac{k \cdot q_1 q_2}{F}$$

$$\therefore r = \sqrt{\frac{k \cdot q_1 q_2}{F}}$$

$$= \sqrt{\frac{9 \times 10^9 \times 26 \times 10^{-6} \times 47 \times 10^{-6}}{5.70}}$$

$$= 1.389 \text{ m}$$

Hence,

$$q_1 = 26 \mu\text{C}$$
$$= 26 \times 10^{-6} \text{ C}$$

$$q_2 = 47 \times 10^{-6} \text{ C}$$

$$F = 5.70 \text{ N}$$

Answers to the Q. No-02

a)

given,

1st acceleration, $a_1 = 7 \text{ m/s}^2$

2nd " $a_2 = 9 \text{ m/s}^2$

Mass of the 1st particle, $m_1 = 6.3 \times 10^{-7} \text{ kg}$

Mass of the 2nd " , $m_2 = ?$

By newton's law,
for particle 1,

$$F = m_1 a_1 \\ = 6.3 \times 10^{-7} \times 7 \\ = 4.41 \times 10^{-6} \text{ N}$$

$\therefore$ Mass of second particle,

$$m_2 = \frac{F}{a_2} \\ = \frac{4.41 \times 10^{-6}}{9} \\ = 4.9 \times 10^{-7} \text{ kg}$$

b)

given,

charge of particle,

$$q_1 = q_2 = q$$

distance, $r = 3.2 \times 10^{-3}$

Applying this to Coulomb's law,

$$F = \frac{k \cdot q_1 \cdot q_2}{r^2}$$

$$\Rightarrow F = \frac{k \cdot q^2}{r^2}$$

$$\Rightarrow q = \sqrt{\frac{F \cdot r^2}{k}}$$

$$= \sqrt{\frac{4.41 \times 10^{-6} \times (3.2 \times 10^{-3})^2}{9 \times 10^9}}$$

$$q = 7.08 \times 10^{-11} \text{ C}$$

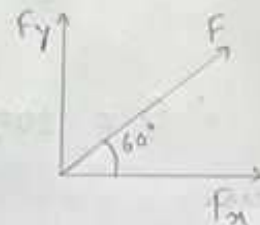
Answer to the q. No-03

i)

$$\therefore 40 = F \cos 60^\circ$$

$$\Rightarrow 40 \times 2 = F$$

$$\therefore F = 80 \text{ N}$$



$$F_x = F \cos 60^\circ$$

$$F_y = F \sin 60^\circ$$

$$\therefore F_y = 80 \sin 60^\circ$$

$$= 80 \times \frac{\sqrt{3}}{2}$$

$$= 40\sqrt{3} \text{ N}$$

ii.)

$$\vec{F}_1 \cdot \vec{F}_2 = |\vec{F}_1| \cdot |\vec{F}_2| \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\vec{F}_1 \cdot \vec{F}_2}{|\vec{F}_1| \cdot |\vec{F}_2|}$$

$$\Rightarrow \cos \theta = \frac{(3\hat{i} - 4\hat{j}) \cdot (-2\hat{i} + 3\hat{k})}{\sqrt{3^2 + (-4)^2} \times \sqrt{(-2)^2 + 3^2}}$$

$$\Rightarrow \theta = \cos^{-1} \left\{ \frac{-6}{\sqrt{25} \times \sqrt{13}} \right\}$$

$$\therefore \theta = 100.44^\circ$$

A.

Hence,

$$\vec{F}_1 = 3\hat{i} - 4\hat{j}$$

$$\vec{F}_2 = -2\hat{i} + 3\hat{k}$$

Answer to the Q.

given,

x component of $\vec{A}$ is 3
y " " " " " " " " " " " "

$$\vec{A} = 3\hat{i} + 4\hat{j}$$

$$= -25\hat{i} + 40\hat{j}$$

$$\therefore |\vec{A}| = \sqrt{(-25)^2 + (40)^2}$$

$$= 47.17 \text{ m}$$

Answers to the Q. No-04

given,

$$x \text{ component of } \vec{A} = -25\text{m}$$

$$y \text{ " " } \vec{A} = 40\text{m}$$

$$a) \vec{A} = x\hat{i} + y\hat{j}$$

$$= -25\hat{i} + 40\hat{j}$$

$$\therefore |\vec{A}| = \sqrt{(-25)^2 + (40)^2}$$

$$= 47.17\text{m}$$

$$b) \vec{A} \cdot \hat{i} = |\vec{A}| |\hat{i}| \cos \theta$$

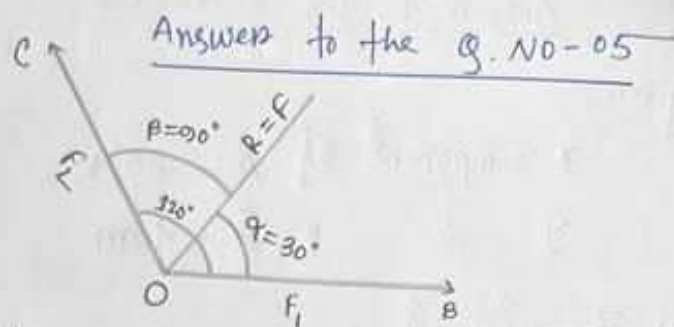
$$\Rightarrow \cos \theta = \frac{\vec{A} \cdot \hat{i}}{|\vec{A}| |\hat{i}|}$$

$$\Rightarrow \cos \theta = \frac{(-25\hat{i} + 40\hat{j}) \cdot \hat{i}}{47.17 \times 1}$$

$$\Rightarrow \theta = \cos^{-1} \left\{ \frac{-25}{47.17} \right\}$$

$$\therefore \theta = 122^\circ$$

Ans.



Let the bigger force be F_1 and the smaller force be F_2 which are acting at angle of 120° .
given that,

$$F_1 = 5N \text{ and } \beta = 90^\circ \text{ and } \alpha + \beta = 120^\circ$$

$$\therefore \alpha = 120^\circ - 90^\circ = 30^\circ$$

First, we find the component of F_1 using the formula,

$$F_1 = \frac{F \sin \beta}{\sin(\alpha + \beta)}$$

$$\Rightarrow F = \frac{F_1 \sin(\alpha + \beta)}{\sin \beta}$$

$$\Rightarrow F = \frac{5 \sin(120^\circ)}{\sin(90^\circ)}$$

$$\therefore F = 4.33 N$$

component F_2 is
formula - $F_2 =$
 $\Rightarrow F_2 =$

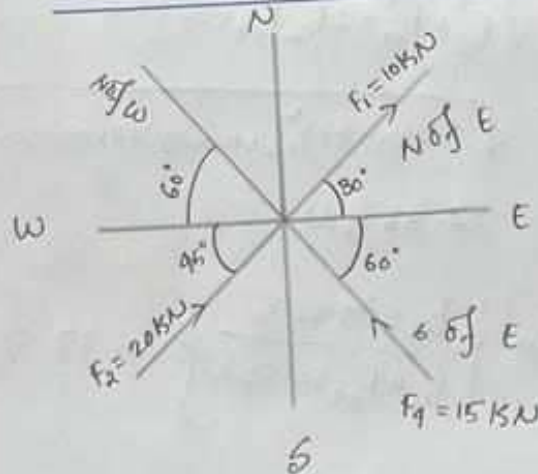
component F_2 can be found using the same formula -

$$F_2 = \frac{F \sin \alpha}{\sin(\alpha + \beta)}$$

$$\Rightarrow F_2 = \frac{4.33 \sin 30^\circ}{\sin 120^\circ}$$

$$\therefore F_2 = 2.5 \text{ N}$$

Answer to the q. NO - 06



F_1 & F_2 -

$$\begin{aligned} F_{12} &= \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos 15^\circ} \\ &= \sqrt{10^2 + 20^2 + 2 \times 10 \times 20 \cos 15^\circ} \\ &= 29.77 \text{ kN} \end{aligned}$$

F_3 & F_4 -

$$\begin{aligned} F_{34} &= \sqrt{F_3^2 + F_4^2 + 2 F_3 F_4 \cos 180^\circ} \\ &= \sqrt{5^2 + 15^2 + 2 \times 5 \times 15 \cos 180^\circ} \\ &= 10 \text{ kN} \end{aligned}$$

$$\begin{aligned} \therefore F_{\text{net}} &= \sqrt{F_{12}^2 + F_{34}^2 + 2 F_{12} F_{34} \cos 82.5^\circ} \\ &= \sqrt{10^2 + (29.77)^2 + 2 \times 10 \times 29.77 \cos 82.5^\circ} \\ &= 32.36 \text{ kN} \end{aligned}$$

$$\theta_{\text{net}} = \tan^{-1} \left\{ \frac{F_{34} \sin 82.5^\circ}{F_{12} + F_{34} \cos 82.5^\circ} \right\} = 17.73^\circ$$

$\therefore \theta_{\text{net}} = 17.73^\circ + 37.5^\circ = 55.23^\circ$ is with + x axis on clockwise.

Answer to the

given, $F = 2 \times 10^{-3} \text{ N}$

Let, $n_1 = n$ on

Let, n be distan

$$f = \frac{K \cdot n_1 \cdot n_2}{n \cdot \lambda}$$

$$\Rightarrow 3 \times 10^{-3} = \frac{K \cdot 20}{n \cdot \lambda}$$

$$\lambda = 2 \times 10^{-3}$$

Answer to the q. No-07

given,

$$F = 2 \times 10^{-3} \text{ N}$$

$$\text{Let, } q_1 = q \text{ and } q_2 = 2q$$

let, r be distance between them.

$$F = \frac{k \cdot q_1 q_2}{r^2}$$

$$\Rightarrow 3 \times 10^{-3} = \frac{k \cdot 2q^2}{r^2}$$

$$\Rightarrow k \cdot 2q^2 = 3 \times 10^{-3} \times r^2$$

$$\Rightarrow k q^2 = 1.5 \times 10^{-3} \times r^2 \quad \text{--- (1)}$$

When r is changes to $(r+0.1) \text{ m}$, force is $5 \times 10^{-4} \text{ N}$

$$\therefore F = \frac{k \cdot q_1 q_2}{(r+0.1)^2}$$

$$\Rightarrow F = \frac{k \cdot 2q^2}{(r+0.1)^2}$$

$$\Rightarrow \frac{5 \times 10^{-4}}{2} = \frac{k q^2}{(r+0.1)^2}$$

$$\Rightarrow 2.5 \times 10^{-4} = \frac{1.5 \times 10^{-3} n^2}{(n+0.1)^2}$$

$$\Rightarrow \frac{2.5 \times 10^{-4}}{1.5 \times 10^3} = \frac{r^2}{(r+0.1)^2}$$

$$\Rightarrow 0.408 = \frac{n}{n+0.1}$$

$$\Rightarrow p = 0.408p + 0.0408$$

$$\Rightarrow r - 0.408r = 0.0408$$

$$\Rightarrow r = \frac{0.0408}{(1 - 0.408)}$$

$$\therefore r_0 = 0.0689 \text{ m}$$

from (i)

$$K \cdot a^v = 1.5 \times 10^{-3} \times 10^v$$

$$\Rightarrow a = \sqrt{\frac{1.5 \times 10^3 \times (0.0689)^2}{a \times 10^9}}$$

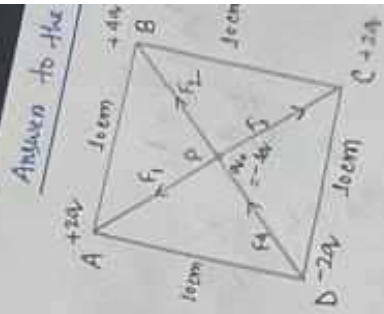
$$\therefore a_v = 2.81 \times 10^{-8} \text{ C}$$

$$\therefore a_1 = 2.81 \times 10^{-8} \text{ C}$$

$$a_2 = 2a_1$$

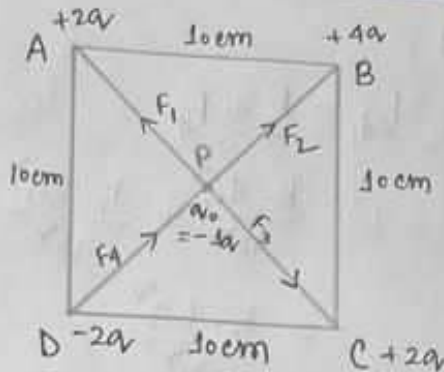
$$\therefore n_2 = 5.61 \times 10^{-8} \text{ C}$$

axis on clockw...



Answers to the q. No-08

i)



ii)

$$F_1 = \frac{k \cdot q_1 \cdot +2q}{r_1^2} = \frac{k \cdot 2q^2}{r_1^2}$$

$$F_2 = k \cdot \frac{-1q \cdot (+4q)}{r_2^2} = \frac{k \cdot 4q^2}{r_2^2}$$

$$F_3 = k \cdot \frac{-1q \cdot (+2q)}{r_3^2} = \frac{k \cdot 2q^2}{r_3^2}$$

$$F_4 = k \cdot \frac{-1q \cdot +2q}{r_4^2} = \frac{k \cdot 2q^2}{r_4^2}$$

$\therefore$ Diagonal of a square, $D = \sqrt{2}a = 0.1\sqrt{2}$

$\therefore$ Radius of square, $r = D/2 = \frac{0.1\sqrt{2}}{2} = 0.07\text{m}$

$$\begin{aligned} \therefore F_1 \text{ and } F_3, \quad F_{13} &= F_3 - F_1 \\ &= \frac{k \cdot 2q^2}{r} - \frac{k \cdot 2q^2}{r} \\ &= 0\text{N} \end{aligned}$$

F_2 and F_4 ,

$$F_{24} = F_2 + F_4$$

$$= \frac{k \cdot 4q^2}{r^2} + \frac{k \cdot 2q^2}{r^2}$$

$$= \frac{k \cdot 6q^2}{r^2}$$

$$\therefore F_{\text{net}} = \sqrt{F_{13}^2 + F_{24}^2}$$

$$= \sqrt{0^2 + F_{24}^2}$$

$$= F_{24} = \frac{k \cdot 6q^2}{r^2}$$

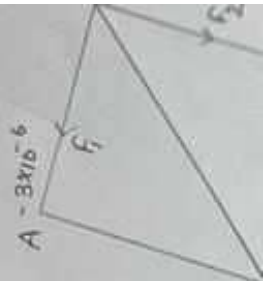
$\therefore$ Direction of Q_{net} is along F_2 .

$$\triangle BCD, \cos \theta = \frac{CD}{BD}$$

$$\Rightarrow \theta = \cos^{-1} \left\{ \frac{0.1}{0.1\sqrt{2}} \right\}$$

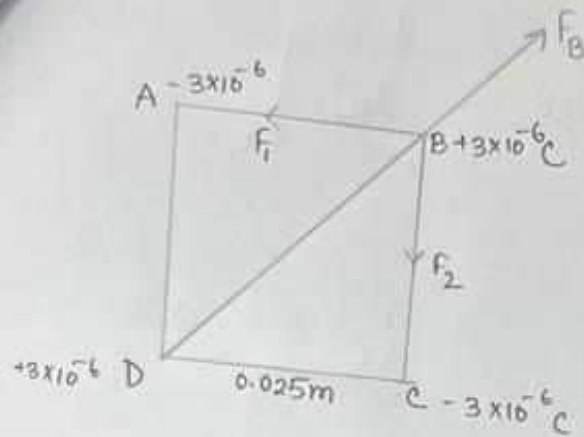
$$= 45^\circ$$

$\therefore Q_{\text{net}} 45^\circ$ along x axis.



Answer to the Q. No-09

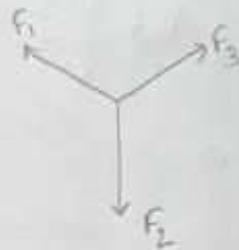
13



Hence,

$$K = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$D = \sqrt{2} \times 0.025 \\ = 0.035 \text{ m}$$



$$F_1 = \frac{K \cdot q_A \cdot q_B}{r^2} = \frac{9 \times 10^9 \times 3 \times 10^{-6} \times 3 \times 10^{-6}}{(0.025)^2} = 120.6 \text{ N}$$

$$F_2 = \frac{K \cdot q_C \cdot q_B}{r^2} = \frac{9 \times 10^9 \times 3 \times 10^{-6} \times 3 \times 10^{-6}}{(0.025)^2} = 120.6 \text{ N}$$

$$F_3 = \frac{K \cdot q_D \cdot q_B}{D^2} = \frac{9 \times 10^9 \times 3 \times 10^{-6} \times 3 \times 10^{-6}}{(0.035)^2} = 66.12 \text{ N}$$

$$\therefore F_{12} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 0^\circ}$$

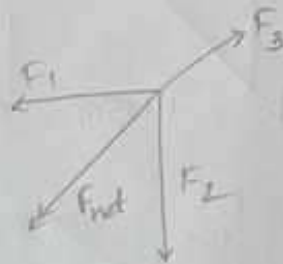
$$= 183.28 \text{ N}$$



$$\therefore F_{\text{net}} = F_{12} - F_3$$

$$= 183.28 - 66.12$$

$$= 117.16 \text{ N}$$



ABED -

$$\therefore \theta_{\text{net}} = \tan^{-1} \left\{ \frac{BC}{DC} \right\}$$

$$= \tan^{-1} \left\{ \frac{129.6}{129.6} \right\}$$

$$= 45^\circ$$

Ans: →

Applying Coulomb

$$F = \frac{k \cdot q_1 \cdot q_2}{r^2}$$

$$\Rightarrow F = \frac{k \cdot q^2}{r^2}$$

$$\therefore q = 2ne^-$$

Answer to the q. No-10

Applying Coulomb's law,

$$F = \frac{k \cdot q_1 \cdot q_2}{r^2}$$

$$\Rightarrow F = \frac{k \cdot q^2}{r^2} \quad \text{--- (i)}$$

$$\therefore q = 2ne^-$$

$$1 \text{ mol Na} = 23 \text{ gm}$$

$$1 \text{ mol Na, } e^- \text{ available} = N_A$$

$$23 \text{ gm Na } e^- \text{ available} = 6.023 \times 10^{23}$$

$$\therefore 1 \text{ gm " " " } = \frac{6.023 \times 10^{23}}{23}$$

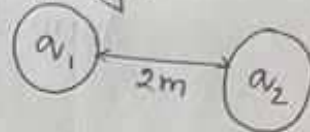
$$\therefore 4 \text{ gm " " " } = \frac{4 \times 6.023 \times 10^{23}}{23}$$

$$\therefore n = 1.047 \times 10^{23}$$

$$\therefore q = 11 \times 1.047 \times 10^{23} \times 1.6 \times 10^{-19}$$

$$= 1.84 \times 10^5 \text{ C}$$

$$m = 4 \text{ gm}$$



$$q_1 = q_2 = q$$

Hence,



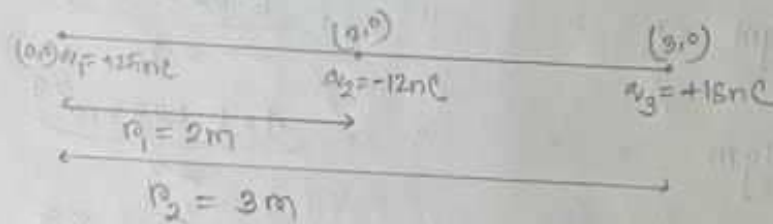
from (i) $\Rightarrow$

$$F = \frac{k \cdot q^2}{r^2}$$

$$= \frac{9 \times 10^9 \times (1.84 \times 10^{-5})^2}{(2)^2}$$

$$\therefore F = 7.61 \times 10^{-10} \text{ N}$$

Answer to the Q. No-11



$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2$$

$$F_1 = k \cdot \frac{q_1 q_2}{r_1^2} = \frac{k \cdot 25 \times -12}{2^2}$$
$$= - \frac{300k}{4}$$

$$F_2 = k \cdot \frac{q_2 q_3}{r_2^2} = k \cdot \frac{25 \times 18}{3^2}$$

$$= \frac{450 \text{ K}}{9}$$

$$\therefore F = F_1 - F_2 = -\frac{300 \text{ K}}{4} + \frac{450 \text{ K}}{9}$$

$$= -\frac{300 \times 9 \times 10^9}{4} + \frac{450 \times 9 \times 10^9}{9}$$

$$= -2.25 \times 10^{-11} \text{ N}$$

$$\theta_{\text{net}} = \tan^{-1} \left\{ \frac{F_2 \sin 180^\circ}{F_1 + F_2 \cos 180^\circ} \right\}$$

$$= \tan^{-1} \left\{ \frac{F_2 \times 0}{F_1 + F_2 \times (-1)} \right\}$$

$$= 0^\circ$$

Answer to the Q. No - 12

given,

$$\text{distance, } r = 5 \text{ nm} = 5 \times 10^{-9} \text{ m}$$

$$G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

$$\text{mass} = m_1 = m_2 = m = 1.67262192 \times 10^{-27} \text{ kg}$$

Electrostatic force,

$$F_E = k \cdot \frac{q_1 \cdot q_2}{r^2}$$

$$= 9 \times 10^9 \times \frac{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(5 \times 10^{-9})^2}$$

$$= 9.21 \times 10^{-12} \text{ N}$$

Gravitational force,

$$F_G = \frac{G \cdot m_1 m_2}{r^2}$$

$$= 6.6743 \times 10^{-11} \times \frac{(1.67262192 \times 10^{-27})^2}{(5 \times 10^{-9})^2}$$

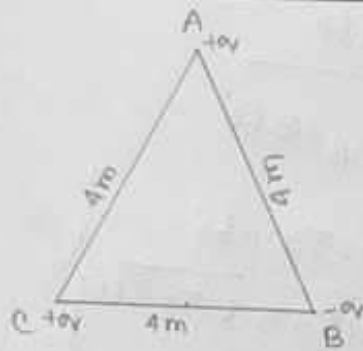
$$= 7.46 \times 10^{-48} \text{ N}$$

$$\therefore \frac{F_E}{F_G} = \frac{9.21 \times 10^{-12}}{7.46 \times 10^{-48}}$$

$$\Rightarrow F_E = 1.23 \times 10^{36} F_G$$

$$\therefore F_E \approx 10^{36} F_G$$

Answers to the Q. No-13



Here,
 $q = 100 \times 10^{-6} \text{ C}$
 $r = 4 \text{ m}$

$$F_{AB} = k \cdot \frac{q_A \cdot q_B}{r^2} = 9 \times 10^9 \times \frac{100 \times 10^{-6} \times 100 \times 10^{-6}}{4^2}$$

$$= 5.625 \text{ N}$$

$$F_{CB} = k \cdot \frac{q_C \cdot q_B}{r^2} = 9 \times 10^9 \times \frac{100 \times 10^{-6} \times 100 \times 10^{-6}}{4^2}$$

$$= 5.625 \text{ N}$$

$$\therefore F = \sqrt{F_{AB}^2 + F_{CB}^2 + 2F_{AB} \cdot F_{CB} \cos 60^\circ}$$

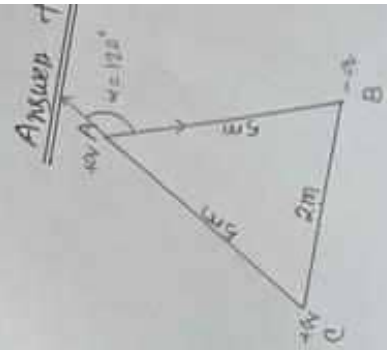
$$= 9.74 \text{ N}$$

$$\therefore \tan \theta = \frac{F_{AB} \sin 60^\circ}{F_{CB} + F_{AB} \cos 60^\circ}$$

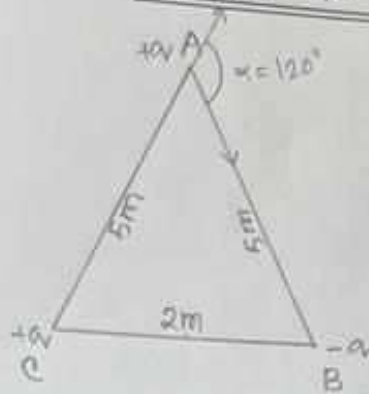
$$\therefore \theta = \tan^{-1} \left\{ \frac{5.625 \sin 60^\circ}{5.625 + 5.625 \cos 60^\circ} \right\}$$

$$= 30^\circ$$

A



Answer to the Q. No-14



Here,

$$q_A = q_B = q_C = 20 \text{ nC} \\ = 20 \times 10^{-9} \text{ C}$$

$$r = r_{CA} = r_{BA} = 5 \text{ m}$$

$$r_{BC} = 2 \text{ m}$$

$$\therefore F_{CA} = k \cdot \frac{q_A q_C}{r^2} = 9 \times 10^9 \times \frac{20 \times 10^{-9} \times 20 \times 10^{-9}}{(5)^2} \\ = 1.44 \times 10^{-7} \text{ N}$$

$$\therefore F_{BA} = k \cdot \frac{q_A q_B}{r^2} = 9 \times 10^9 \times \frac{20 \times 10^{-9} \times 20 \times 10^{-9}}{(5)^2} \\ = 1.44 \times 10^{-7} \text{ N}$$

The magnitude of the total force on the charge at A,

$$F = \sqrt{F_{CA}^2 + F_{BA}^2 + 2 F_{CA} \cdot F_{BA} \cos 120}$$

$$\Rightarrow F = \sqrt{F_x^2 + F_y^2} = F \cdot F$$

$$\Rightarrow F = \sqrt{F^2}$$

$$\therefore F = 1.44 \times 10^{-7} \text{ N}$$

$\therefore$ direction,

$$\tan \theta = \frac{F_{CA} \sin 120^\circ}{F_{BA} + F_{CA} \cos 120^\circ}$$

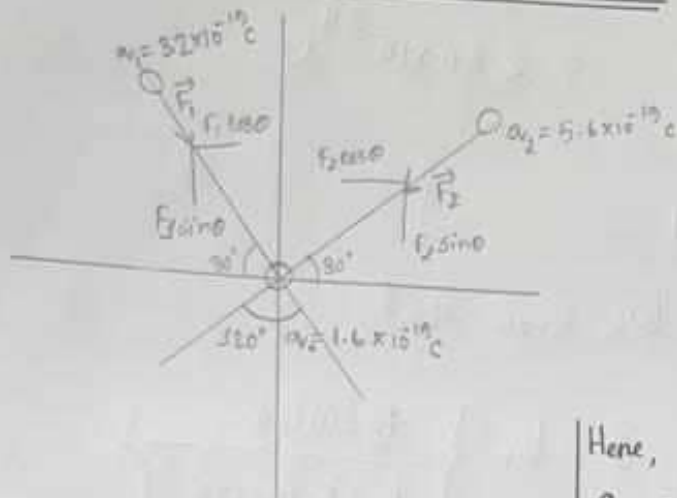
$$\Rightarrow \theta = \tan^{-1} \left\{ \frac{1.44 \times 10^{-7} \times \sin 120^\circ}{1.44 \times 10^{-7} + 1.44 \times 10^{-7} \cos 120^\circ} \right\}$$

$$\therefore \theta = 60^\circ$$

A

Answer to the Q. NO-15

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23



a)

$$\begin{aligned} \therefore F_1 &= k \cdot \frac{q_1 q_0}{r^2} \\ &= 9 \times 10^9 \times \frac{3.2 \times 10^{-19} \times 1.6 \times 10^{-19}}{(0.005)^2} \\ &= 1.84 \times 10^{-23} \text{ N} \end{aligned}$$

Here,

$$\begin{aligned} q_1 &= 3.2 \times 10^{-19} \text{ C} \\ q_2 &= 5.6 \times 10^{-19} \text{ C} \\ r_1 &= 0.005 \text{ m} \\ q_0 &= 1.6 \times 10^{-19} \text{ C} \\ \alpha &= 120^\circ \end{aligned}$$

$$\begin{aligned} \therefore F_2 &= k \cdot \frac{q_2 q_0}{r^2} = 9 \times 10^9 \times \frac{5.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(0.005)^2} \\ &= 3.22 \times 10^{-23} \text{ N} \end{aligned}$$

$\therefore$ Magnitude of net force on q_0 ,

$$\begin{aligned} F_{\text{net}} &= \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos 120^\circ} \\ &= \sqrt{(1.84 \times 10^{-23})^2 + (3.22 \times 10^{-23})^2 + 2 \times 1.84 \times 10^{-23} \times 3.22 \times 10^{-23} \times \cos 120^\circ} \end{aligned}$$

$$= 2.80 \times 10^{-23} \text{ N}$$

b)

We know that,

$$\theta = \tan^{-1} \left\{ \frac{F_2 \sin 120^\circ}{F_1 + F_2 \cos 120^\circ} \right\}$$

$$= \tan^{-1} \left\{ \frac{3.22 \times 10^{-23} \times \sin 120^\circ}{1.84 \times 10^{-23} + 3.22 \times 10^{-23} \times \cos 120^\circ} \right\}$$

$$= 85.28^\circ$$

A.

The magnitude
force is,

Answer to the Q. No-18

$$\therefore F_{\text{net}} = -F_1 + F_2 - F_3$$

$$\Rightarrow 0 = -k \cdot \frac{q_1 q_2}{r_1^2} + k \cdot \frac{q_2 q_3}{r_2^2} - k \cdot \frac{q_1 q_3}{r_3^2}$$

$$\Rightarrow 0 = kq \left\{ -\frac{q_1}{r_1^2} + \frac{q_2}{r_2^2} - \frac{q_3}{r_3^2} \right\}$$

$$\Rightarrow 0 = -\frac{8}{8^2} + \frac{5}{(14)^2} - \frac{q_3}{(26)^2}$$

$$\Rightarrow 0 = -\frac{39}{392} - \frac{q_3}{(26)^2}$$

$$\Rightarrow \frac{q_3}{(26)^2} = -\frac{39}{392}$$

$$\therefore q_3 = -\frac{39}{392} \times (26)^2$$

$$= -67.25 \mu\text{C}$$

Ans

Here,

$$r_1 = 8\text{m}$$

$$r_2 = 14\text{m}$$

$$r_3 = 26\text{m}$$

$$q_1 = +8 \mu\text{C}$$

$$q_2 = -5 \mu\text{C}$$

$$q_3 = ?$$

We know,

$$F = k \cdot \frac{q_1 q_2}{r^2}$$

Answers to the q. No-19

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We know,

$$F = \frac{k \cdot q_1 \cdot q_2}{r^2}$$

$$\Rightarrow r^2 = \frac{k \cdot q^2}{F}$$

$$\therefore r = \sqrt{\frac{k \cdot q^2}{mg}}$$

$$= \sqrt{\frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{1.67 \times 10^{-19} \times 9.8}}$$

$$= 0.119 \text{ m}$$

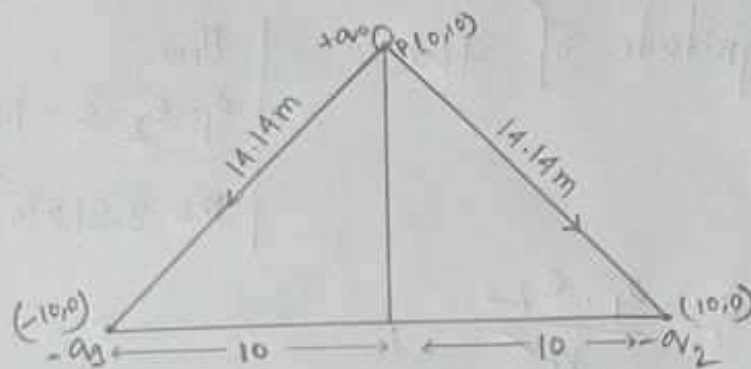
Ans.

Here,

$$q_1 = q_2 = q = 1.6 \times 10^{-19} \text{ C}$$

$$m_p = 1.67 \times 10^{-19} \text{ kg}$$

Answer to the Q. No-17



a)

$$F_1 = k \cdot \frac{q_0 \cdot q_1}{r^2}$$

$$= 9 \times 10^9 \times \frac{10 \times 10^{-5} \times 10 \times 10^{-5}}{(14.14)^2}$$

$$= 0.45 \text{ N}$$

Here,

$$q_0 = q_1 = q_2$$

$$q = 10 \times 10^{-5}$$

$$r = \sqrt{10^2 + 10^2}$$

$$= 14.14 \text{ m}$$

$$F_2 = k \cdot \frac{q_0 \cdot q_2}{r^2} = 9 \times 10^9 \times \frac{(10 \times 10^{-5})^2}{(14.14)^2}$$

$$= 0.45 \text{ N}$$

$$\therefore F_{\text{net}} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 60^\circ}$$

$$= \sqrt{(0.45)^2 + (0.45)^2 + 2 \times (0.45)^2 \cos 60^\circ}$$

$$= 0.78 \text{ N}$$

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27
∴ direction of the force,

$$\theta = \tan^{-1} \left\{ \frac{F_2 \sin 60^\circ}{F_1 + F_2 \cos 60^\circ} \right\}$$

$$= \tan^{-1} \left\{ \frac{0.45 \sin 60^\circ}{0.45 + 0.45 \cos 60^\circ} \right\}$$

$$\theta = 30^\circ$$

$$\therefore \theta_{\text{net}} = 30^\circ$$

Ans:

Answer to the Q. No-16

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The magnitude of the force is,

Here,

$$q_1 = q_2 = e = 1.6 \times 10^{-19} \text{ C}$$

$$r = 3.81 \times 10^{-7} \text{ m}$$

$$F = k \cdot \frac{q_1 \cdot q_2}{r^2}$$

$$= k \cdot \frac{e^2}{r^2}$$

$$= 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{(3.81 \times 10^{-7})^2}$$

$$= 1.58 \times 10^{-5} \text{ N}$$